

FX_Rotation_Nomura_LeenBouAlia

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1 FX Trading Rotation

1.0.1 1.1 Currency Pair Structure

EUR/USD = 1.0850

This means **1 Euro costs 1.0850 US Dollars**.

- **Base currency** (left): what you are buying or selling

- **Quote currency** (right): what you pay or receive

1.0.2 1.2 The Hierarchy Rule (ISO Priority)

Market convention ranks currencies such that:

Priority	Currency	Mnemonic
1st	XAU / XAG (Gold, Silver)	Commodities first
2nd	EUR	Europe
3rd	GBP	Great Britain
4th	AUD	Australia
5th	NZD	New Zealand
6th	USD	United States
7th	CAD, CHF, JPY, others	Rest of world

So EUR/USD, not USD/EUR. GBP/JPY, not JPY/GBP.

Pairs not involving USD are called **cross rates** (e.g. EUR/JPY, GBP/CHF).

1.0.3 1.3 Bid / Ask / Spread

Term	Definition
Bid	Price dealer <i>buys</i> base (you sell base)
Ask / Offer	Price dealer <i>sells</i> base (you buy base)
Spread	Ask – Bid = dealer's compensation
Mid	(Bid + Ask) / 2 — used for mark-to-market

EUR/USD: Bid 1.0848 / Ask 1.0852 → spread = 0.0004 = 4 pips

1.0.4 1.4 Pips and Figures

- **Pip** = the 4th decimal place for most pairs (0.0001)
- **Figure / Big figure** = the first 3 decimals (e.g. “1.08”)
- **JPY pairs** are quoted to 2 decimal places $\rightarrow$ 1 pip = 0.01

USD/JPY: 149.52 $\rightarrow$ 149.57 = +5 pips (not +0.0005)

1.0.5 1.5 Cross Rate Calculation

To get EUR/JPY when you only have EUR/USD and USD/JPY:

$$\text{EUR/JPY} = \text{EUR/USD} \times \text{USD/JPY}$$

$$= 1.0850 \times 149.50 = 162.21$$

General rule: **multiply if USD cancels in the middle; divide if it doesn't.**

1.0.6 Market Context: What Drives Spot FX?

Driver	Effect on USD	Intuition
Fed hikes rates	USD strengthens	Higher yield attracts capital
Risk-off sentiment	USD strengthens	Safe-haven demand
Current account surplus	Domestic FX strengthens	Export receipts converted
Inflation surprise (US)	USD weakens (initially)	Real rate erosion
ECB hikes unexpectedly	EUR/USD rises	Euro rates more attractive

1.1 2. FX Products Overview

1.1.1 2.1 Spot

Definition: Exchange of two currencies at the current market rate, settled in **T+2 business days** (T+1 for USD/CAD, same-day for some).

Use case: Immediate currency conversion — e.g. a European exporter receiving USD revenue converts to EUR.

Payoff (long EUR/USD spot at S_0 , closed at S_T):

$$P\&L = (S_T - S_0) \times \text{Notional}$$

If you believe EUR will appreciate: Buy EUR/USD spot.

Entry: 1.0850. If EUR rallies to 1.1050 $\rightarrow$ +200 pips per unit notional.

1.1.2 2.2 FX Forwards & Outrights

Definition: An agreement to exchange currencies at a **pre-agreed rate (forward rate F)** on a **specific future date**, entered today with no upfront payment.

The forward rate is not a forecast — it is **mechanically derived** from spot and the interest rate differential:

$$F = S_0 \times \frac{1 + r_{\text{quote}} \cdot T}{1 + r_{\text{base}} \cdot T}$$

Where: - S_0 = spot rate - r_{quote} = interest rate of quote currency (e.g. USD) - r_{base} = interest rate of base currency (e.g. EUR) - T = time to maturity in years

Example (EUR/USD, 1-year):

$$F = 1.0850 \times \frac{1 + 0.053}{1 + 0.038} = 1.0850 \times 1.0145 = 1.1007$$

USD rates are higher than EUR rates $\rightarrow$ forward USD is *cheaper* $\rightarrow$ EUR/USD forward trades **above spot (forward premium on EUR)**.

Forward Points In practice, traders quote the **forward points** (pips added to spot):

$$\text{Forward Points} = F - S_0 = 1.1007 - 1.0850 = +157 \text{ pips}$$

Sign	Meaning
Positive points	Base currency at premium (base country lower rates)
Negative points	Base currency at discount (base country higher rates)

Payoff

$$P\&L_{\text{long forward}} = (S_T - F) \times \text{Notional}$$

No premium paid — but you are **obligated** to transact at F .

Trade: Believe EUR will appreciate above the forward rate of 1.1007 by year-end $\rightarrow$ buy EUR/USD 1Y forward.

If spot at expiry is 1.1200 $\rightarrow$ profit = $(1.1200 - 1.1007) \times \text{notional}$.

If spot is 1.0800 $\rightarrow$ **loss** = $(1.0800 - 1.1007) \times \text{notional}$. No optionality.

1.1.3 2.3 FX Swaps

Definition: A simultaneous **spot purchase + forward sale** (or vice versa) of the same currency pair and notional. Two legs, agreed at inception.

$$\text{FX Swap} = \underbrace{\text{Buy spot at } S_0}_{\text{near leg}} + \underbrace{\text{Sell forward at } F}_{\text{far leg}}$$

This is NOT a currency swap (which involves exchanging interest payments).

Purpose: Roll a forward position, manage funding gaps, or swap currency for a fixed period without taking spot risk.

The swap price = the forward points only (the spot rate nets out).

Practical use: A bank needs EUR for 3 months but has USD. It buys EUR spot and simultaneously sells EUR 3M forward — locking in both legs at known rates.

1.1.4 2.4 FX Options

Term	Definition
Call	Right to <i>buy</i> base currency at strike K
Put	Right to <i>sell</i> base currency at strike K
Premium	Upfront cost (unlike forwards)
European	Exercise only at expiry (standard in FX)
American	Exercise any time (less common in FX)

Payoff (long EUR call / USD put, strike K):

$$\text{Payoff} = \max(S_T - K, 0) \times \text{Notional}$$

FX Option Greeks

Greek	Symbol	Meaning in FX context
Delta	Δ	USD P&L per 1-pip move in spot; also probability ITM
Gamma	Γ	How fast delta changes; spikes near expiry ATM
Vega		P&L per 1-vol-point move in implied vol
Theta	Θ	Daily time decay cost of holding the option
Rho		Sensitivity to interest rate differential — more important in FX than in equity options

Note on Rho in FX: Because FX options have *two* interest rates (domestic and foreign), they carry **two Rhos** — one for each rate. This connects directly to the interest rate parity section below.

Common FX Option Structures

Structure	Construction	View
Risk Reversal	Long OTM call + Short OTM put (same delta)	Bullish on base, captures skew
Straddle	Long ATM call + Long ATM put	Long vol, direction-neutral
Strangle	Long OTM call + Long OTM put	Cheaper long-vol play
Butterfly	Long 2 ATM + Short 2 wings	Short vol, range-bound
Seagull	Long call + Short higher call + Short lower put	Structured hedge with zero/low cost

If you believe EUR will appreciate and want leveraged exposure:

Buy a 1-month 25-delta EUR call (OTM). Low premium, high leverage.

If EUR/USD moves from 1.0850 $\rightarrow$ 1.1050, the 25 Δ call may double or triple in value.

Risk: Lose entire premium if EUR stays flat or falls.

1.2 3. Interest Rate Parity (IRP)

Interest rate parity is the **fundamental no-arbitrage constraint** linking spot FX rates, forward rates, and interest rate differentials across two countries.

1.2.1 3.1 Covered Interest Rate Parity (CIP)

The core idea: You should earn the same return whether you invest in domestic currency directly, or convert to foreign currency, invest there, and lock in the exchange rate with a forward.

$$F = S_0 \times \frac{1 + r_d}{1 + r_f}$$

Or equivalently (for small rate differences):

$$\frac{F - S_0}{S_0} \approx r_d - r_f$$

If CIP is violated $\rightarrow$ riskless arbitrage exists:

Scenario	Action
$F > S_0(1 + r_d)/(1 + r_f)$	Borrow domestic, convert to foreign, invest, lock forward $\rightarrow$ riskless profit
$F < S_0(1 + r_d)/(1 + r_f)$	Borrow foreign, convert, invest domestic, lock forward $\rightarrow$ riskless profit

Pre-2008: CIP held almost perfectly.

Post-2008: CIP *persistently* breaks down — this is the **FX basis** (covered in Section 5).

1.2.2 3.2 Uncovered Interest Rate Parity (UIP)

The idea: Without a forward to lock in rates, the expected spot rate should adjust so that both currencies offer the same *expected* return.

$$\mathbb{E}[S_T] = S_0 \times \frac{1 + r_d}{1 + r_f}$$

High-yielding currency should *depreciate* by the rate differential to prevent free returns.

Reality: UIP fails empirically — high-yielding currencies tend to *appreciate*, not depreciate. This is the **carry trade anomaly**.

The Carry Trade

Step	Action
1	Borrow in low-rate currency (e.g. JPY at 0.1%)
2	Convert to high-rate currency (e.g. USD at 5.3%)
3	Invest at higher rate
4	Pocket the rate differential (~5.2%) if spot doesn't move against you

Risk: A sudden reversal — if JPY appreciates sharply (carry unwind), losses on the FX position can exceed the interest carry. This is called **carry crash risk** and is associated with spikes in FX implied volatility.

Carry trade P&L = $(r_{\text{high}} - r_{\text{low}}) \times T - \Delta S$ (spot move against position)

Current context (2025-2026): With Fed rates at ~5.25% and BOJ rates still near 0.5%, USD/JPY carry remains attractive but increasingly unstable as the BOJ normalises policy. Watch for vol spikes as an early warning.

1.3 5. Implied vs. Realised Vol — and the Analogy to FX Rates

This section explores a conceptual bridge that is not standard in most textbooks but is increasingly used by sophisticated practitioners: **the parallel between the vol surface in equity/FX options and the interest rate term structure in fixed income — and how both “transfer” into FX.**

1.3.1 5.1 Recap: Implied vs. Realised Vol in Options

Concept	Definition	Market signal
Implied Vol (IV)	Vol backed out of option prices via Black-Scholes; the market's <i>expectation</i> of future realised vol + a risk premium	Reflects demand for options, fear, hedging pressure
Realised Vol (RV)	Actual historical vol of returns over a period	What actually happened
Vol Risk Premium (VRP)	$IV - RV > 0$ on average	Options are expensive on average; sellers earn a premium for providing insurance

The **vol surface** (IV across strikes and tenors) encodes: - **Skew**: puts are more expensive than calls (demand for downside protection) → negative skew in equities - **Term structure**: near-term IV vs. long-term IV; usually upward sloping in calm markets, inverts during stress

1.3.2 5.2 The Interest Rate Analogy

Now consider the **interest rate world**:

Rate Concept	Parallel to Vol
Spot rate / overnight rate	Realised vol (what is actually happening now)
Forward rate	Implied vol for a future period
Rate term structure	Vol term structure (shape matters: flat, upward, inverted)
Rate risk premium	Vol risk premium (forwards/implied embed a premium over expected)

Just as implied vol $>$ realised vol on average (sellers earn the VRP), **forward rates** $>$ **realised future spot rates** on average (the expectations hypothesis fails; term premium exists).

1.3.3 5.3 How This Transfers Into FX — The FX Basis as “Rate Implied Vol”

Here is the key insight:

The FX basis is to interest rates what the vol risk premium is to volatility.

Break it down:

In Volatility:

- You observe an **implied vol** from option prices
- The market is pricing in expected realised vol + **a risk premium**
- When risk aversion rises → IV spikes above RV → the gap (VRP) widens

In FX / Rates:

- You observe an **implied foreign interest rate** derived from FX swap prices (via CIP)
- This implied rate should equal the actual foreign rate — but it doesn't
- The gap is the **FX basis** — a risk premium for dollar funding access
- When funding stress rises → basis widens → the gap between implied USD rate and actual USD rate grows

$$\underbrace{IV - RV}_{\text{Vol Risk Premium}} \quad \longleftrightarrow \quad \underbrace{r_{USD}^{\text{implied via FX}} - r_{USD}^{\text{actual}}}_{\text{FX Basis (negative sign convention)}}$$

Both premiums: - Are **structurally positive/negative** (not zero) - **Widen during stress** (risk-off, liquidity crises) - Are **mean-reverting** but can persist for extended periods - Are **harvested by sellers** in normal times (option sellers earn VRP; FX swap providers earn basis)

1.3.4 5.4 Implied Rates from FX Markets

Given spot S_0 , forward F , and domestic rate r_d , you can **back out the implied foreign rate**:

$$r_f^{\text{implied}} = \left(\frac{S_0}{F} \right) (1 + r_d) - 1$$

This is exactly what the FX swap market does when it prices the basis. If:

$$r_f^{\text{implied}} \neq r_f^{\text{actual}}$$

The difference is the **basis** — the market's “implied interest rate” diverges from the real rate, just as IV diverges from RV.

1.3.5 5.5 Visualising the Parallel

VOL WORLD

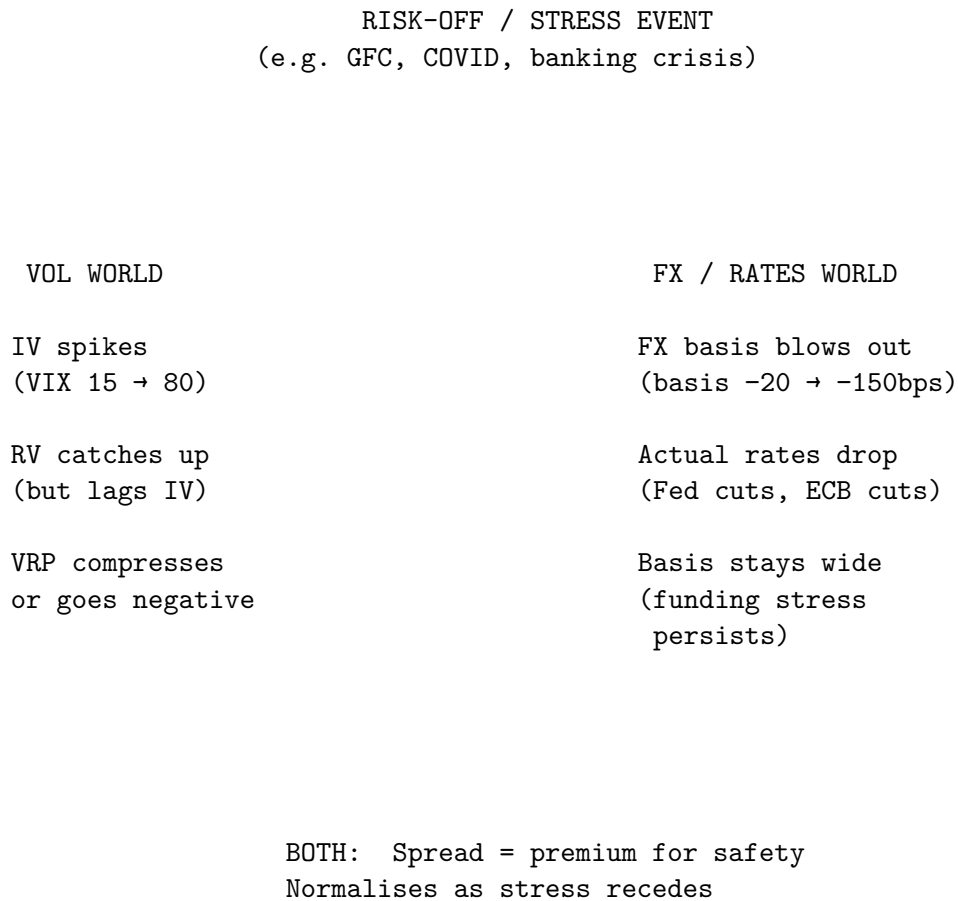
Implied Vol (IV)
Realised Vol (RV)
Vol Risk Premium (IV - RV)
Vol term structure
Vol skew (puts > calls)
Gamma (short-dated IV spikes)
VIX spike on risk-off
Calendar spread: near vs far
Option seller earns VRP

FX / RATES WORLD

Implied foreign rate (from FX)
Actual policy/LIBOR/SOFR rate
FX Basis (CIP deviation)
Yield curve / rate term structure
Basis skew (USD premium in stress)
Short-end basis spikes at quarter-end
Basis blow-out on risk-off (-150bps)
FX swap: near leg vs far leg
FX swap provider earns basis spread

1.3.6 5.6 Market Conditions Driving Each

Diagram enhanced with AI



1.3.7 5.7 Trade Implications

Scenario: You believe the Fed will hold rates while ECB cuts

Market implication	Direction
USD rates stay elevated relative to EUR	EUR/USD forward points widen
Implied EUR rate (from FX) falls	EUR/USD basis could move more negative
EUR/USD spot likely weakens	USD strength
EUR vol skew may shift	Puts on EUR/USD get more expensive

Trades: 1. **Sell EUR/USD forward** — benefit from widening forward premium (USD rates — EUR rates) 2. **Buy EUR/USD put** (long USD call) — directional option play on EUR weakness 3. **Receive EUR/USD basis swap** — earn the basis if you can fund in EUR and swap to USD

Scenario: Sudden risk-off, dollar squeeze

Signal	Trade
Basis blows wide negative	Buy USD/JPY put (JPY safe-haven appreciation)
IV spikes in short-dated FX options	Sell gamma (if stress fades quickly) or buy near-term straddles
Term structure inverts (near IV > far IV)	Long front-month vol vs short back-month vol (reverse calendar in FX vol)

Key insight: The same reverse calendar spread logic you applied in equity vol applies directly here — when near-term FX IV spikes above long-term IV (vol term structure inverts), a reverse calendar captures the dislocation.

1.4 6. Quick Reference: FX Products Summary Table

Product	Settlement	Premium?	Obligation?	Primary Use
Spot	T+2	No	Yes	Immediate conversion
Outright Forward	Future date	No	Yes	Hedge known future flow
FX Swap	Two legs (near + far)	No	Yes	Roll/fund positions
XCCY Basis Swap	Periodic + maturity	No	Yes	Long-term funding in foreign currency
Vanilla Option	Expiry date	Yes	No (buyer)	Directional or vol play
Risk Reversal	Expiry date	Low/zero cost	Partial	Hedged directional view
Straddle	Expiry date	Yes	No	Long vol, no direction

1.5 7. Key Formulas Cheat Sheet

$$\text{Forward Rate: } F = S_0 \cdot \frac{1 + r_d \cdot T}{1 + r_f \cdot T}$$

$$\text{Forward Points: } F - S_0 \approx S_0 \cdot (r_d - r_f) \cdot T$$

$$\text{Implied Foreign Rate: } r_f^{\text{implied}} = \frac{S_0}{F}(1 + r_d) - 1$$

$$\text{FX Basis: } \delta = r_f^{\text{implied}} - r_f^{\text{actual}}$$

$$\text{Carry P\&L: } \pi = (r_{\text{high}} - r_{\text{low}}) \cdot T - \Delta S_{\text{adverse}}$$

$$\text{BSM FX Call (Garman-Kohlhagen): } C = S_0 e^{-r_f T} N(d_1) - K e^{-r_d T} N(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r_d - r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Notes compiled for Nomura FX Desk rotation · Last updated June 2026

1.6 FX Swap

1.7 Spot Risk in FX Swaps

1.8 FX Basis

1.9 Cross Currency Swaps (XCCY)

1.10 The Cross Currency Basis

1.11 Fixed-to-Fixed and Fixed-to-Floating Swaps

1.12 Relative Value